VOLKSWAGEN

A 7-Slide Business & Data Analytics Case Study

Mobility & Brands

Operations

Supply Chain

Data & Decisions

Case-study lens: how a global automotive group creates value, manages complexity, and uses data.

Understanding the Volkswagen Group

Founded

Volkswagen was established in 1937. Today, the Group operates a large portfolio of automotive brands.

Global footprint

A multinational automotive group with manufacturing, sales and service activities across major world markets.

Brand portfolio

Volkswagen Group includes brands such as Volkswagen Passenger Cars, Audi, Škoda, SEAT/CUPRA, Porsche, Bentley, Lamborghini and others.

Core business: designing, manufacturing and selling passenger cars, commercial vehicles and mobility-related services.

Strategic themes: electrification, software, connected vehicles, premium mobility and operational efficiency.

Key challenge: coordinate many brands and markets while responding quickly to changing customer preferences.

From engineering to customer experience

Product & R&D

Design, engineering, software and new technologies.

Manufacturing

Factories, automation, quality control and production planning.

Supply chain

Parts, suppliers, logistics, inventory and procurement.

Sales & mobility

Dealers, digital channels, financing and mobility services.

Customer

Ownership, service, retention and brand experience.

Analytics can connect the entire chain: demand forecasting → production planning → inventory → sales → after-sales service.

The case-study problem

Electrification

Shift from conventional powertrains toward EVs while maintaining profitability and customer choice.

Software & connectivity

Vehicles increasingly behave like software platforms, raising expectations around updates, features and digital services.

Supply-chain complexity

Global suppliers, semiconductor availability, logistics and inventory require resilient planning.

Sustainability

Reduce emissions across vehicles, factories and supply chains while meeting evolving regulations.

Case question

How can better data help Volkswagen improve demand planning, production, quality and customer experience?

Turning vehicle and operations data into decisions

Demand forecasting

Predict model-level demand by market, season and customer segment.

Production analytics

Compare planned vs. actual output, downtime, throughput and quality.

Inventory optimization

Balance tyres, parts and finished vehicles against expected demand.

Quality analytics

Detect recurring defects, warranty patterns and service issues.

Logistics analytics

Monitor delivery lead times, route performance and bottlenecks.

Customer analytics

Understand features, service behavior, retention and satisfaction.

A practical analyst view

Units Sold

Track volume by model/market

Revenue

Monitor sales value & growth

Inventory Days

Detect over/under-stock

OEE

Measure manufacturing effectiveness

Dashboard pages: Executive Overview • Sales • Production • Supply Chain • Quality • Customer.

Suggested drill-downs: Country → Region → Dealer → Model → Variant.

Core stack for a learner: Excel for exploration → SQL for querying → Power BI for dashboards → Python for advanced analysis.

Decision example: if demand rises but inventory days fall below target, planners can prioritize production or supplier allocation.

🚗 Volkswagen's scale creates both opportunity and complexity.

📊 Data analytics can connect customer demand with production, supply chain, quality and financial outcomes.

🔧 The strongest dashboards do more than report numbers—they highlight exceptions and support decisions.

▯ For a data analyst, a useful portfolio project is to build a simulated Volkswagen sales + inventory dashboard using SQL and Power BI.

🏁 Final takeaway

Better decisions come from connecting the right data to the right business question.

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DATA → INSIGHT → ACTION